

Math 118 – Lecture 03

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September 8, 2026

1 Orthonormal

Definition 1.1 (Orthogonal Complement). If V_0 is a subspace of V , the orthogonal complement $V_0^\perp$ of V_0 is

$$V_0^\perp = \{u \in V : \langle u, v \rangle = 0 \forall v \in V_0\}$$

Why is $V_0^\perp$ a subspace?

If

$$x, y \in V_0^\perp, \alpha \in K \leftarrow (\text{field of scalars: } \mathbb{R} \text{ or } \mathbb{C}) \text{ and } v \in V_0$$

Then, you have that

$$(\alpha x + y, v) = \alpha(x, v) + (y, v) = 0$$

because each of these terms is 0.

remember gram-schmidt

Definition 1.2 (Orthogonal Projection). If $e_1, \dots, e_N$ is an orthonormal basis for a subspace V_0 of V , the *orthogonal projection* of V onto V_0 is the map $P : V \rightarrow V_0$ defined by

$$P(u) = \sum_{n=1}^N \langle u, e_n \rangle e_n$$

Read this as the “shadow” of u on V_0 : keep exactly the e_n -components of u and discard the rest. The properties that make P useful:

Proposition 1.3 (Basic properties of P) *Let $e_1, \dots, e_N$ be an orthonormal basis of a subspace $V_0 \leq V$, and let P be as defined above. Then:*

1. (**Linearity**) $P(\alpha u + v) = \alpha Pu + Pv$ for all $u, v \in V$ and $\alpha \in \mathbb{C}$.
2. (**Identity on V_0**) $Pe_m = e_m$ for every m ; in particular $Pu = u$ for $u \in V_0$, $Pu = 0$ for $u \in V_0^\perp$, and $P^2 = P$.
3. (**Residual is orthogonal**) $u - Pu \perp v_0$ for every $u \in V$ and every $v_0 \in V_0$.

Proof. (1) By linearity of the inner product in its first slot,

$$\begin{aligned} P(\alpha u + v) &= \sum_{n=1}^N \langle \alpha u + v, e_n \rangle e_n = \sum_{n=1}^N (\alpha \langle u, e_n \rangle + \langle v, e_n \rangle) e_n \\ &= \alpha \sum_{n=1}^N \langle u, e_n \rangle e_n + \sum_{n=1}^N \langle v, e_n \rangle e_n = \alpha Pu + Pv. \end{aligned}$$

(2) $Pe_m = \sum_n \langle e_m, e_n \rangle e_n = e_m$ since $\langle e_m, e_n \rangle = \delta_{mn}$. If $u \in V_0$, expand $u = \sum_n \langle u, e_n \rangle e_n$ (coefficients by inner products), so $Pu = u$; if $u \in V_0^\perp$ then every $\langle u, e_n \rangle = 0$, so $Pu = 0$. Finally $P^2u = P(Pu) = Pu$ because $Pu \in V_0$ and P is the identity there. (3) First check the residual against each basis vector: since $Pu = \sum_n \langle u, e_n \rangle e_n$,

$$\langle Pu, e_m \rangle = \sum_n \langle u, e_n \rangle \underbrace{\langle e_n, e_m \rangle}_{\delta_{nm}} = \langle u, e_m \rangle, \quad \text{so} \quad \langle u - Pu, e_m \rangle = 0 \text{ for every } m.$$

Now take any $v_0 \in V_0$ and expand $v_0 = \sum_m \langle v_0, e_m \rangle e_m$; by conjugate-linearity in the second slot,

$$\langle u - Pu, v_0 \rangle = \sum_m \overline{\langle v_0, e_m \rangle} \langle u - Pu, e_m \rangle = 0. \quad \square$$

Remark 1.4 (Decomposition). Every $u \in V$ splits as $u = Pu + (u - Pu)$ with $Pu \in V_0$ and $u - Pu \in V_0^\perp$, so $V = V_0 \oplus V_0^\perp$.

Theorem 1.5 (Closest Point Property) *For every $u \in V$ and every $v_0 \in V_0$,*

$$\|u - Pu\| \leq \|u - v_0\|,$$

with equality if and only if $v_0 = Pu$. So Pu is the unique closest point of V_0 to u .

Proof. Fix $v_0 \in V_0$ and decompose

$$u - v_0 = \underbrace{(u - Pu)}_{\in V_0^\perp} + \underbrace{(Pu - v_0)}_{\in V_0}.$$

The two pieces are orthogonal by property (3), so Pythagoras gives

$$\|u - v_0\|^2 = \|u - Pu\|^2 + \|Pu - v_0\|^2 \geq \|u - Pu\|^2,$$

and taking square roots gives the inequality. Equality forces $\|Pu - v_0\| = 0$, i.e. $v_0 = Pu$. (This is the same Pythagoras proved in the Lecture 02 proofs section.) $\square$

Geometric significance: Pu is the foot of the perpendicular dropped from u onto V_0 – in $\mathbb{R}^3$, the closest point on a plane to a point off the plane, which is exactly the “shadow” picture. The residual $u - Pu$ is the error you cannot remove: any other candidate $v_0 \in V_0$ differs from Pu by a vector *inside* V_0 , which is perpendicular to that error and can only add to it (Pythagoras). This is

the least-squares idea, and it is why, later in the course, Fourier partial sums will be the best L^2 approximations by trigonometric polynomials: they are projections.

Theorem 1.6 (Variational Characterization) *Let $u \in V$ and $u_0 \in V_0$. Then*

$$\langle u - u_0, v_0 \rangle = 0 \text{ for all } v_0 \in V_0 \iff \|u - u_0\| = \min_{v_0 \in V_0} \|u - v_0\|.$$

That is, “the residual is orthogonal to V_0 ” is not merely analogous to “ u_0 is the closest point” – the two statements are equivalent, and both hold exactly when $u_0 = Pu$.

Proof. ($\Rightarrow$) Assume $\langle u - u_0, v_0 \rangle = 0$ for all $v_0 \in V_0$. For any $v_0 \in V_0$, write $u - v_0 = (u - u_0) + (u_0 - v_0)$; the first piece is orthogonal to V_0 by hypothesis and the second lies in V_0 , so Pythagoras gives

$$\|u - v_0\|^2 = \|u - u_0\|^2 + \|u_0 - v_0\|^2 \geq \|u - u_0\|^2.$$

Thus u_0 attains the minimum, uniquely (equality forces $v_0 = u_0$).

($\Leftarrow$) Assume u_0 minimizes: $\|u - u_0\| \leq \|u - v_0\|$ for all $v_0 \in V_0$. Fix $v_0 \in V_0$ with $v_0 \neq 0$ and a real t ; since V_0 is a subspace, $u_0 + tv_0 \in V_0$, so the function

$$g(t) = \|u - (u_0 + tv_0)\|^2 = \|u - u_0\|^2 - 2t \operatorname{Re}\langle u - u_0, v_0 \rangle + t^2 \|v_0\|^2$$

has a minimum at $t = 0$. Hence $g'(0) = -2 \operatorname{Re}\langle u - u_0, v_0 \rangle = 0$. Running the same argument with $iv_0 \in V_0$ gives $0 = \operatorname{Re}\langle u - u_0, iv_0 \rangle = \operatorname{Im}\langle u - u_0, v_0 \rangle$. Real and imaginary parts both vanish, so $\langle u - u_0, v_0 \rangle = 0$ for all $v_0 \in V_0$. $\square$

Why this matters: the variational side is how closest points are actually *computed* – instead of guessing the best approximation, you solve the orthogonality conditions $\langle u - u_0, v_0 \rangle = 0$ for all $v_0 \in V_0$ (the *normal equations*). B&N §0.5.2 and the least-squares material in §0.7 are exactly this equivalence in action.

2 Linear Operators

Let V, W be vector spaces over K . $T : V \rightarrow W$ is linear if $T(u + v) = T(u) + T(v)$. Examples of linear operators :

- $A : C^n \rightarrow C^m$ (matrix) where $(Ax)_i = \sum_{j=1}^n A_{ij}x_j$ ($1 \leq i \leq m$)
- $G : l^2(a, b) \rightarrow l^2(a, b) = Gf(x) = \int_a^b G(x, y)f(y)dy$

Another fundamental example. If $\phi_1, \dots, \phi_N$ are linearly independent vectors in V , define $T : \mathbb{C}^N \rightarrow V$ by

$$Tc = \sum_{n=1}^N c_n \phi_n$$

(T is linear: $T(\alpha c + d) = \sum_n (\alpha c_n + d_n) \phi_n = \alpha Tc + Td$.)

Proposition 2.1 (Range of T) *The range of T is exactly the subspace of V spanned by the ϕ_n :*

$$\text{range}(T) = \left\{ \sum_{n=1}^N c_n \phi_n : c \in \mathbb{C}^N \right\} = \text{span}\{\phi_1, \dots, \phi_N\}.$$

Since the ϕ_n are linearly independent, T is also one-to-one, so T is an isomorphism from $\mathbb{C}^N$ onto its range.

Why: every Tc is by definition a linear combination of the ϕ_n , so $\text{range}(T) \subseteq \text{span}\{\phi_1, \dots, \phi_N\}$; conversely, every vector in the span has the form $\sum_n c_n \phi_n = Tc$ for some $c \in \mathbb{C}^N$, so the two sets are equal. For one-to-one: if $Tc = Td$ then $\sum_n (c_n - d_n) \phi_n = 0$, so linear independence forces $c_n = d_n$ for every n , i.e. $c = d$.

Example 2.2 (Polynomials in $L^2(0, 1)$). Take $V = L^2(0, 1)$, $N = 4$, and

$$\phi_1 = 1, \quad \phi_2 = x, \quad \phi_3 = x^2, \quad \phi_4 = x^3.$$

These are linearly independent in $L^2(0, 1)$: if $a_1 + a_2x + a_3x^2 + a_4x^3 = 0$ as an element of L^2 (i.e. except on a set of measure zero), the polynomial vanishes at infinitely many points of $(0, 1)$, so it is the zero polynomial and $a_1 = a_2 = a_3 = a_4 = 0$.

Here the operator is

$$Tc = c_1 + c_2x + c_3x^2 + c_4x^3, \quad c = (c_1, c_2, c_3, c_4) \in \mathbb{C}^4,$$

and by theorem 2.1 its range is

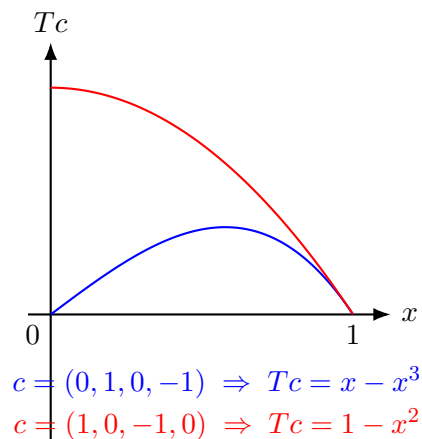
$$\text{span}\{1, x, x^2, x^3\} = P_3 = \{\text{all polynomials of degree } \leq 3\},$$

a 4-dimensional subspace of $L^2(0, 1)$.

With a choice of $c \in \mathbb{C}^4$ you get a concrete polynomial:

- $c = (0, 1, 0, -1)$: $Tc = x - x^3$
- $c = (1, 0, -1, 0)$: $Tc = 1 - x^2$
- $c = (2, -1, 3, 1)$: $Tc = 2 - x + 3x^2 + x^3$

Conversely, every polynomial in P_3 is Tc for a *unique* $c \in \mathbb{C}^4$, because the ϕ_n are independent and T is one-to-one: e.g. $p(x) = 1 + x - 2x^3$ corresponds to $c = (1, 1, 0, -2)$.



Definition 2.3 (Linear functional). A linear functional on V is a linear operator whose target vector space is the scalars themselves, $\mathbb{C}$ (or $\mathbb{R}$):

$$l : V \rightarrow \mathbb{C}, \quad l(\alpha u + \beta v) = \alpha l(u) + \beta l(v).$$

Examples of linear functionals:

- On $\mathbb{C}^n$: $l(x) = \sum_{j=1}^n a_j x_j$ for a fixed vector $a \in \mathbb{C}^n$; e.g. $l(x) = x_1 + 2x_3$.
- The most important example: for a fixed $y \in V$,

$$l(v) = \langle v, y \rangle$$

is a linear functional, because the inner product is linear in its first slot. On $L^2(0, 1)$ this reads $l(f) = \int_0^1 f(t) \overline{g(t)} dt$ for a fixed $g \in L^2(0, 1)$.

- Special case on $L^2(0, 1)$ with $g = 1$: $l(f) = \int_0^1 f(t) dt$, the average value of f – its constant (“DC”) Fourier coefficient.

Remark 2.4. Not every linear functional is visibly an inner product, but on the spaces we care about (Hilbert spaces) every bounded linear functional has this form: $l(v) = \langle v, y \rangle$ for a unique $y \in V$. This fact is the Riesz representation theorem.

Definition 2.5 (Dual Space). The space of all linear functionals on V is itself a vector space, called the *dual space* of V :

$$V^* = \{l : V \rightarrow \mathbb{C} : l \text{ linear}\}, \quad (\alpha l + \beta m)(v) = \alpha l(v) + \beta m(v).$$

Why is V^* a vector space? The pointwise operations inherit every axiom from the scalars; for instance additivity,

$$(\alpha l + \beta m)(u + v) = \alpha l(u + v) + \beta m(u + v) = (\alpha l + \beta m)(u) + (\alpha l + \beta m)(v),$$

and the zero functional $l(v) \equiv 0$ is the additive identity.

On $\mathbb{C}^n$ the dual space is n -dimensional: every functional is $l(x) = \sum_{j=1}^n a_j x_j$, a combination of the coordinate functionals $l_j(x) = x_j$. In a Hilbert space the same thing happens through the inner product: by the Riesz remark above, $y \mapsto \langle \cdot, y \rangle$ matches each $y \in V$ with a distinct functional and exhausts all of (bounded) V^* – so the dual space is a copy of V itself.

Definition 2.6 (Convolution). The convolution of g and f is

$$(g * f)(x) = \int_{-\infty}^{\infty} g(x - y) f(y) dy$$

This is an integral operator as above, with kernel $G(x, y) = g(x - y)$.

Find the solution of the boundary value problem

$$\begin{cases} -u''(x) = f(x), & 0 < x < 1, \\ u(0) = 0, \\ u(1) = 0. \end{cases}$$

Show that the solution can be written in the form

$$u(x) = \int_0^1 G(x, y) f(y) dy,$$

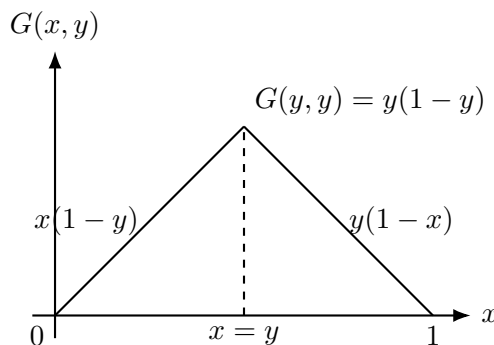
where $G(x, y)$ is the Green's function for the differential operator

$$L = -\frac{d^2}{dx^2}$$

with the boundary conditions

$$G(0, y) = G(1, y) = 0.$$

Derive the Green's function $G(x, y)$.



$$G(x, y) = \begin{cases} x(1 - y), & 0 \leq x \leq y, \\ y(1 - x), & y \leq x \leq 1. \end{cases}$$

Definition 2.7 (kernel). The kernel $\ker(G)$ = the null space of G also written as $N(G)$ its nullspace

3 Topics to review

Green's functions and the continuous (L^2) applications of vector spaces are the parts of this lecture that still need work. When going back over these notes, review in this order.

Green's function material

- **The problem being solved:** $-u''(x) = f(x)$ on $0 < x < 1$ with $u(0) = u(1) = 0$ – why it is reasonable to look for a solution of the form $u(x) = \int_0^1 G(x, y)f(y) dy$ at all.
- **Deriving $G(x, y)$:** away from $x = y$ the equation is $-G''(\cdot, y) = 0$, so G is linear in x on each side. The two conditions that pin it down are *continuity* at $x = y$ (where $G(y, y) = y(1 - y)$) and the *jump condition* on the derivative,

$$G_x(y^+, y) - G_x(y^-, y) = -1,$$

which comes from integrating $-u'' = f$ across $x = y$. Re-derive the piecewise formula ($x(1 - y)$ for $x \leq y$, $y(1 - x)$ for $y \leq x$) from just these two conditions plus $G(0, y) = G(1, y) = 0$.

- **Verifying the solution:** differentiate $u(x) = \int_0^1 G(x, y)f(y) dy$ under the integral separately on $x < y$ and $x > y$, and see exactly where the jump in G_x produces $u'' = -f$; then check $u(0) = u(1) = 0$ directly.
- **Kernel viewpoint:** G is the kernel of an integral operator $Gf(x) = \int_0^1 G(x, y)f(y) dy$ on $L^2(0, 1)$ – compare with the convolution kernel $G(x, y) = g(x - y)$ in the Convolution definition above.

Continuous vector spaces and operators on them

- **Functions as vectors:** $L^2(0, 1)$ as the infinite-dimensional analogue of $\mathbb{C}^n$ – vectors are functions, inner products are integrals, and convergence in L^2 is *neither* pointwise *nor* uniform convergence in general (B&N §0.3.2; the single most important nuance for the rest of the course).
- **Finite-dimensional subspaces inside L^2 :** the polynomial example ($T : \mathbb{C}^4 \rightarrow L^2(0, 1)$ with range P_3); how orthogonal complements and projections (§0.5.2) will let us decompose functions the way we decompose vectors.
- **Linear operators on function spaces:** the matrix example versus the integral operator example (§0.6.1) – what it takes for $Gf(x) = \int_a^b G(x, y)f(y) dy$ to be a well-defined map of L^2 into itself.
- **Functionals and the dual space:** functionals are operators with scalar output; the inner product functionals $v \mapsto \langle v, y \rangle$, and (Riesz) why on Hilbert spaces they exhaust the bounded dual.
- **Convolution:** $(g * f)(x) = \int g(x - y)f(y) dy$ as the integral operator whose kernel depends only on the difference $x - y$; it returns as the main character of the Fourier transform chapter (B&N §2.2.2, p. 107).

Textbook anchors and extra problems for this review

- Read: B&N §0.3 (especially §0.3.2, convergence in L^2 versus uniform), §0.5.2 (orthogonal complements/projections), §0.6.1 (linear operators). Adjoints (§0.6.2) are the likely next lecture.
- Chapter 0 exercises (pp. 34–38) beyond the homework: #2 (verify the l^2 and L^2 inner products of Examples 0.2–0.3), #4 (L^2 inner-product axioms, including positivity via continuity), #5 (inner product on l^2), and especially #6, #7, #8 – sequences that converge in L^2 but not uniformly or pointwise.